Hierarchical Shell-Manifold Theory: Operator Closures, Relational Geometry, and the Boundary of Derived Physics

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Abstract

Hierarchical Shell-Manifold Theory (HSMT) is a graded operator framework in which an integer hierarchy, an associative operator laboratory, localized inter-grade coupling, and projected effective dynamics are used to organize candidate emergent physics. Earlier presentations emphasized the breadth of the programme. The present work instead records the outcome of a post-maturation closure programme designed to determine what follows from the declared mathematical structures before observed physics is used as a target.

We separate every substantive claim into one of six statuses: derived, conditional, primitive, no-go/nonselection, open, or historically demoted. Within this discipline, several positive structures survive: controlled Feshbach reduction, relational locality and metric-measure-energy layers, projective/Hodge and Clifford seeds, internal-symmetry commutant structure, family and flavor carrier spaces, scalar and neutral-mass mechanisms, locality-preserving continuum relabeling limits, an intrinsic relabeling-invariant spectral geometric energy, and a gapped harmonic determinant with a controlled low-frequency response and positive induced collective inertia.

The same analysis establishes substantial negative results. Complete exchangeability does not select physical locality; grade does not select dimension or time; Feshbach projection does not select the observed sector; Clifford structure does not select a physical fermion sector or Standard Model gauge group; family multiplicity does not select three generations; free overlap models do not predict flavor data; scalar kinematics does not derive a Higgs vacuum; neutral-mass algebra does not select the neutrino branch; emergent metric structure does not imply gravity; chart covariance does not imply gravitational diffeomorphism symmetry; and the current background-free harmonic relational gravity benchmark possesses no nontrivial unconstrained vacuum and no regulator-universal continuum kernel.

The resulting theory is therefore more constrained than the original broad unification proposal but also more explicit about its present limits. HSMT does not currently derive the observed spacetime dimension, Standard Model content, symmetry-breaking vacuum, neutrino mass branch, or Einstein gravity. Its current strongest results are the operator architecture, several conditional mechanisms, relational geometric constructions, and a sharply localized set of missing microscopic inputs required for physical closure.

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Notation guide

Symbol	Meaning
$\mathcal{M} = \bigsqcup_N \mathcal{M}_N$	Graded HSMT arena.
N	Integer hierarchy grade; not a physical dimension.
d	Generic dimensional observable or effective dimension.
$\rho_N = N - N_*$	Grade coordinate after choice of additive origin.
$L0$	Primitive locality filtration.
$ME0$	Primitive metric-measure-energy completion.
$T0$	Primitive affine clock/generator structure.
P, Q	Chosen visible-sector projector and complement.
$\Sigma(z)$	Feshbach self-energy.
$A(q)$	Declared microscopic fast quadratic operator.
$E_0[q]$	Inherited harmonic spectral vacuum energy, $\frac{1}{2}\operatorname{Tr}\sqrt{A(q)}$.
Γ_{fast}	Fast-sector determinant effective action.
$M^{(\text{ind})}$	Induced collective inertia tensor.
SDiff	Measure/volume-preserving diffeomorphism group, or its relevant local continuum analogue.
`[D]`, `[C]`, `[P]`, `[NG]`, `[OPEN]`, `[DEMOTED]`	Epistemic-status labels defined in Section 2.

1 Introduction

Hierarchical Shell-Manifold Theory was introduced as an attempt to organize physics through a graded geometric arena and an operator-theoretic hierarchy rather than beginning with a fixed low-energy spacetime and particle content. Its central architectural idea is that the observed sector may be an effective projection of a larger graded system whose neighboring grades are not exactly decoupled.

The scientific problem addressed in the present paper is different from the one emphasized in the earliest HSMT drafts. The earlier question was approximately:

> Can a graded shell-manifold architecture be extended so that known low-energy physics can be represented?

The post-maturation question is stricter:

> What follows from the declared HSMT structures before observed physics is used as an upstream target?

That change in question is consequential. A mathematical framework may be capable of hosting a phenomenon without selecting it. A mechanism may be compatible with observed physics without predicting the corresponding physical sector. An operator may possess the correct algebraic form without being operationally identified with the observable world.

The maturation programme was therefore organized around explicit closure tests, counterexamples, and no-go results. Three historical stages should be distinguished.

1. **Broad unification stage.** Early HSMT drafts explored possible links between the grade hierarchy, effective dimensions, internal algebras, particle structure, cosmology, and gravity.
2. **September reference framework.** The September 2026 paper fixed the principal architecture: a graded arena, an associative operator laboratory, localized hierarchy coupling, exact projection, and a deliberately separated downstream extension programme.
3. **Post-maturation closure programme.** Subsequent work systematically separated derived statements from conditional mechanisms, primitive inputs, failed selectors, and open problems.

The present paper is not a rewrite of the September framework. It is a theorem-boundary paper: a record of what survived the closure programme and what did not.

2 Epistemic status language

Every central claim in this paper is assigned a local status.

Table 2: Epistemic status vocabulary used throughout the paper.

Status	Meaning
[D]	Derived mathematically from explicitly declared assumptions.
[C]	Conditional consequence requiring additional named hypotheses.
[P]	Primitive, empirical, operational, or externally supplied input.
[NG]	No-go, nonselection theorem, counterexample, or closed failed selector.
[OPEN]	Well-posed unresolved problem.
[DEMOTED]	Historical claim no longer retained as a present result.

Model theorem versus physical theorem.
A result may be mathematically exact inside a declared HSMT realization without establishing that Nature realizes that realization. Examples include

$$q_I(4) = 2, \qquad E_0 = \frac{1}{2}\operatorname{Tr}\sqrt{A}, \qquad M^{(\mathrm{ind})} \geq 0.$$

These are genuine mathematical or model results. A physical theorem additionally requires the unresolved operational and constitutive bridges identifying the relevant HSMT variables with physical observables. Throughout this paper,

$$\text{model theorem} \neq \text{physical-selection theorem}.$$

Project-wide rule. A closed selector search is not reopened merely because a desired physical result remains missing. Reopening requires new microscopic input, a new theorem, new data, or an identified mathematical error.

3 Frozen architectural spine

The public HSMT architecture remains

$$\mathcal{M} = \bigsqcup_N \mathcal{M}_N \longrightarrow D_\rho, D_{\mathrm{sp}} \longrightarrow \text{states/packets} \longrightarrow \langle N|M\rangle \longrightarrow \mathcal{D} \longrightarrow P_N \longrightarrow H_{\mathrm{eff}}. \tag{1}$$

The measurement layer remains conceptually separate:

$$Q_N(\ell) \longrightarrow d_N(\ell) = \frac{d \ln Q_N(\ell)}{d \ln \ell}. \tag{2}$$

Equation (1) is an architecture. It does not by itself identify the physical grade, physical dimension, physical projector, matter content, or gravitational dynamics.

3.1 Grade is not dimension

[D] The hierarchy label N is an integer structural label.

[NG] There is no theorem equating N with a physical dimension.

[NG] A shell is not assumed to be a conventional spatial shell.

The matured language is therefore

$$\text{shell} \neq \text{dimension}, \qquad N \neq d. \tag{3}$$

Several dimension concepts must remain distinct:

$$d_{\mathrm{geom}}, \qquad d_{\mathrm{res}}, \qquad d_{\mathrm{spec}}, \qquad d_{\mathrm{dyn}}. \tag{4}$$

Equality among them is not assumed.

3.2 Absolute grade origin

[D] The additive origin of grade is conventional unless a physical selector is supplied. A hierarchy coordinate can therefore be written

$$\rho_N = N - N_*. \tag{5}$$

The historical choice $N_* = 4$ is a reference fixing. It is not a derivation of an absolute fourth physical grade.

3.3 Successor semantics

[D] The statement $\Delta N = 1$ means one primitive successor step in the hierarchy.

[OPEN] No unique microscopic ontology of that successor step is presently derived. Support rank is one conditional realization; Krylov depth is another contact-relative construction. Neither is presently a universal physical identification.

4 Associative operator laboratory and projected dynamics

One of the robust parts of HSMT is its associative operator layer. Let the Hilbert space be decomposed into a chosen visible sector and its complement,

$$\mathcal{H} = P\mathcal{H} \oplus Q\mathcal{H}, \qquad Q = I - P, \tag{6}$$

and let a self-adjoint operator be represented in block form,

$$H = \begin{pmatrix} A & V^\dagger \\ V & C \end{pmatrix}. \tag{7}$$

For complex z in the resolvent domain of C, the exact Schur/Feshbach reduction [4, 5] gives

$$H_{\text{eff}}(z) = A + V^\dagger (z - C)^{-1} V. \tag{8}$$

Define

$$\Sigma(z) = V^\dagger (z - C)^{-1} V. \tag{9}$$

Proposition 4.1 (Isolated-sector dressing bound)**.** *Let*

$$H = \begin{pmatrix} A & V^\dagger \\ V & C \end{pmatrix}$$

be self-adjoint and let $z_0 \in \mathbb{R}$ lie a distance at least $\Delta > 0$ from $\mathrm{Spec}(C)$*. Define*

$$R_0 = (z_0 - C)^{-1}, \qquad \Sigma(z) = V^\dagger (z - C)^{-1} V.$$

Then

$$\Sigma'(z_0) = -V^\dagger R_0^2 V \leq 0,$$

and

$$Z = I - \Sigma'(z_0) = I + V^\dagger R_0^2 V \geq I.$$

Moreover,

$$\|Z - I\| \leq \frac{\|V\|^2}{\Delta^2}.$$

Proof. Since C is self-adjoint and z_0 lies outside its spectrum, R_0 is self-adjoint and

$$\frac{d}{dz}(z - C)^{-1} = -(z - C)^{-2}.$$

Hence $\Sigma'(z_0) = -V^\dagger R_0^2 V$. For every ψ,

$$\langle \psi, V^\dagger R_0^2 V \psi \rangle = \|R_0 V \psi\|^2 \geq 0.$$

Therefore $Z \geq I$. Finally, $\|R_0\| \leq \Delta^{-1}$, giving

$$\|Z - I\| \leq \|V\|^2 \|R_0\|^2 \leq \frac{\|V\|^2}{\Delta^2}.$$

□

[D] Proposition 4.1 is an operator theorem once the block decomposition is given. It controls dressing; it does not select the physical projector.

[NG] Exactness of Feshbach reduction does not select P. Spectral isolation also does not generically select it, since a spectral projector may make the off-block coupling vanish by construction.

Projection/memory formalisms provide wider context for such reductions [6, 7, 8].

4.1 Physical-sector bridge

The mature physical-projector package requires at least:

1. operational preparations and readouts;
2. a contact map between laboratory observables and HSMT variables;
3. an intertwiner into the HSMT state space;
4. a projector correspondence;
5. consistency across multiple contacts;
6. a controlled relation between bare and dressed sectors.

[P] These operational data remain outside the frozen mathematical core.

5 Locality and physical geometry

5.1 Why locality must be supplied

The prelocal complete-support constructions possess large permutation symmetry.

[NG] Complete exchangeability does not select a distinguished deterministic local neighborhood structure. A fully equivariant internal rule cannot break a completely symmetric configuration into a unique physical locality structure without additional information.

Accordingly, HSMT now treats locality through a primitive filtration $L0$.

[P] $L0$ supplies nested, relabeling-covariant neighborhood structure. It does not supply cardinal lengths, a physical measure, conductances, diffusion normalization, or physical dimension.

5.2 Metric-measure-energy completion

A physical local geometry needs more than adjacency. The minimal mature completion is a metric-measure-energy package,

$$(d, \mu, \mathcal{E}), \tag{10}$$

or, in a discrete realization,

$$(\ell_e, \mu_v, c_e). \tag{11}$$

[P] This package is called $ME0$. Dirichlet-form and spectral-graph frameworks provide mathematical context for such structures [10, 11].

The three ingredients are logically independent in the general theory. Edge length does not uniquely determine conductance; conductance does not uniquely determine vertex measure; effective resistance need not equal the path metric.

[NG] $L0$ does not derive $ME0$.

5.3 Unit gauge

Constant rescalings of the absolute unit system may modify length, measure, or energy normalizations while leaving logarithmic scaling exponents unchanged.

[D] Dimensional exponents are therefore insensitive to pure global unit calibration. This does not imply that arbitrary changes of conductance at fixed values of other dimensional parameters are gauge.

6 Dimension as response, not grade

Once $ME0$ is given, several scale-dependent exponents can be defined. For a volume function,

$$d_V(\ell) = \frac{d \ln V(\ell)}{d \ln \ell}. \tag{12}$$

For a return probability,

$$d_S(\tau) = -2 \frac{d \ln P(\tau)}{d \ln \tau}. \tag{13}$$

Analogous definitions apply to walk and resistance exponents.

[D] These observables are well-defined relative to the chosen physical geometry.

[NG] No current theorem forces them to coincide.

Scale-dependent dimensionality has also been studied in other frameworks; comparison is contextual and does not imply equivalence with HSMT [13].

Theorem 6.1 (Bounded-distortion stability of scaling exponents)**.** *Let two positive scale observables $Q_1(\ell)$ and $Q_2(\ell)$ satisfy*

$$c_- Q_1(\ell) \le Q_2(\ell) \le c_+ Q_1(\ell)$$

through an asymptotic scaling window, with constants $0 < c_- \le c_+ < \infty$. If Q_1 has asymptotic power-law exponent d, and the multiplicative distortion remains asymptotically bounded without introducing a new power of ℓ, then Q_2 has the same asymptotic exponent.

Proof. Write $Q_2(\ell) = a(\ell) Q_1(\ell)$, where $a(\ell)$ is bounded above and below by positive constants. Then

$$\frac{\ln Q_2(\ell)}{\ln \ell} = \frac{\ln Q_1(\ell)}{\ln \ell} + \frac{\ln a(\ell)}{\ln \ell}.$$

In an asymptotic scaling regime in which $|\ln \ell| \to \infty$, boundedness of a implies $\ln a(\ell)/\ln \ell \to 0$. Equivalent logarithmic-derivative formulations require the corresponding mild regularity of $a(\ell)$. □

Corollary 6.2. *A genuine change of asymptotic dimensional exponent across grades requires an unbounded or scale-dependent distortion, a change in measure/energy/time scaling, new local degrees of freedom, a change in topology/locality class, or a change in the observable being measured.*

[C] Corollary 6.2 is conditional on the existence of the corresponding asymptotic scaling windows. No numerical physical dimension follows.

7 Support geometry, projective structure, and Hodge theory

Proposition 7.1 (Complete-support interface dimension)**.** *For the declared complete-support tangent decomposition, the interface-sector dimension is*

$$q_I(m) = \frac{m(m-3)}{2}.$$

Consequently $q_I(m) = 0$ *for* $m \leq 3$*, while* $q_I(4) = 2$*. Thus* $m = 4$ *is the first rank at which a nontrivial interface sector appears.*

Proof. The complete support possesses $m(m-1)/2$ independent symmetric edge variables. Removing the m independent row-strength directions leaves

$$\frac{m(m-1)}{2} - m = \frac{m(m-3)}{2}$$

interface directions in the declared decomposition. □

[NG] Proposition 7.1 identifies a local combinatorial onset. It is not a derivation of a physical spacetime dimension.

7.1 Relational incidence

Let

$$(B_m x)_{ij} = x_j - x_i. \tag{14}$$

Then

$$B_m(x + cu_m) = B_m x, \tag{15}$$

with $u_m = (1, \ldots, 1)^T$. For connected complete support,

$$\ker B_m = \operatorname{span}(u_m). \tag{16}$$

Thus the common vertex mode may be quotiented as a relational redundancy while retaining nontrivial edge cochains.

A canonical complete-support Dirichlet form has

$$L_m = w(mI - J) = wmP_m, \tag{17}$$

where P_m projects onto the sum-zero relational subspace. Hence the nonzero relational scale behaves as $\lambda_m \sim wm$.

[D] This is a model theorem for the declared complete-support realization.

[OPEN] Which support/cochain sector is physically realized remains unresolved.

8 Time, causal structure, and arrow

HSMT now separates three logically distinct problems.

8.1 Time

[P] A physical affine clock t and self-adjoint generator are supplied in the $T0$ package.

[NG] Grade is not time.

[NG] Feshbach memory presupposes a time variable; it does not derive one.

8.2 Causal structure

For a local effective mode with principal symbol

$$p(\omega, k) = -A\omega^2 + 2B^i \omega k_i + C^{ij} k_i k_j, \tag{18}$$

a sufficient Lorentzian condition is $A > 0$ together with positivity of

$$C^{ij} + \frac{B^i B^j}{A}. \tag{19}$$

[C] Under appropriate hyperbolicity, stability, propagation, and low-energy universality assumptions, a Lorentzian characteristic cone may emerge.

[OPEN] A physical universal cone is not yet derived from the frozen core.

8.3 Arrow

[NG] Closed reversible unitary dynamics does not generically possess a nonconstant globally monotone reduced observable. Grade currents can reverse; entanglement can rise and fall; retarded versus advanced Green functions require boundary/orientation information.

[P] Future orientation and statistical arrow data therefore remain separate $O0/A0$ inputs.

9 Clifford and fermionic kinematic capacity

Let E be a positive inner-product space. Exterior multiplication $\varepsilon(v)$ and adjoint contraction $\iota(v)$ satisfy

$$\{\varepsilon(u), \varepsilon(v)\} = 0, \tag{20}$$
$$\{\iota(u), \iota(v)\} = 0, \tag{21}$$
$$\{\varepsilon(u), \iota(v)\} = \langle u, v \rangle. \tag{22}$$

Thus $c(v) = \varepsilon(v) + \iota(v)$ gives a Clifford representation. For a Hodge complex,

$$D_H = d + d^\dagger, \qquad D_H^2 = dd^\dagger + d^\dagger d = \Delta_H. \tag{23}$$

[D] The algebraic Clifford/Hodge seed is genuine once the metric and cochain structure are supplied.

[NG] This does not by itself derive a physical spinor bundle. Differential forms are generally larger than an irreducible spinor module; support CAR algebra is not automatically physical exchange

statistics; a physical spin or spin^c structure is additional global information in the smooth setting; and a unique first-order Dirac direction is not selected by full permutation symmetry alone. Standard spin-geometry context is provided by [12].

$$\text{Clifford capacity} \neq \text{physical fermion derivation.} \tag{24}$$

10 Internal symmetry: possibility space versus selection

Suppose the finite complex Clifford module decomposes as

$$S \cong \bigoplus_{\alpha} (S_\alpha \otimes \mathbb{C}^{m_\alpha}) . \tag{25}$$

Then

$$\mathrm{Comm}(c) \cong \bigoplus_{\alpha} (I_{S_\alpha} \otimes M_{m_\alpha}(\mathbb{C})) . \tag{26}$$

Its unitary part contains factors of the form $\prod_\alpha U(m_\alpha)$, before additional reality, grading, determinant, or tensor constraints are imposed.

[D] This identifies a mathematically natural home for internal unitary symmetry.

[NG] It does not select the physical gauge group. Neither Clifford algebra alone nor composition-algebra analogies uniquely derive $SU(3) \times SU(2) \times U(1)$, its global quotient, hypercharges, matter representations, or couplings.

[P] These belong to the physical GI0 package unless future microscopic physics selects them.

Noncommutative/spectral geometry provides relevant mathematical comparison where spectral triples or commutants are discussed [9]; no equivalence with HSMT is implied.

11 Families and flavor

11.1 Family multiplicity

A replicated matter module may be represented as

$$\mathcal{H}_{\text{matter}} \cong \mathcal{H}_1 \otimes F, \qquad F \cong \mathbb{C}^r. \tag{27}$$

[D] Family multiplicity is mathematically distinguishable from gauge multiplicity.

[NG] No current theorem selects $r = 3$. K4 interface counting, cycle rank, neighboring grades, triality, generic spectral degeneracy, and anomaly cancellation do not provide a unique family-number selector.

11.2 Flavor overlap mechanism

A general overlap form may be written

$$Y_f = \kappa_f B_{L,f}^\dagger K_f B_{R,f}. \tag{28}$$

For a same-profile positive kernel this reduces to a Gram-type structure. The observable singular values follow from

$$H_L = YY^\dagger, \qquad H_R = Y^\dagger Y. \tag{29}$$

[C] Localized profiles and Gaussian-type kernels can support hierarchical matrix elements.

[NG] With sufficiently free profiles and kernels, such constructions can realize too broad a family of matrices to be predictive.

[NG] A real, even Gaussian does not provide an intrinsic CP-violating phase.

Consequently observed fermion masses, CKM mixing, PMNS mixing, and CP invariants remain downstream of missing physical flavor dynamics.

12 Scalar sector

A 0-cochain or 0-form is a natural scalar-valued object on a local geometry.

[D] HSMT therefore possesses scalar kinematic capacity.

[NG] A geometric scalar is not automatically a physical Higgs field.

A physical symmetry-breaking scalar requires an internal representation R_Φ, kinetic normalization, a physical effective potential, an instability mechanism, stabilizing nonlinear structure, a selected vacuum orbit, and operational identification.

A conventional radial example is

$$V(\Phi) = m_\Phi^2 \langle \Phi, \Phi \rangle + \lambda \langle \Phi, \Phi \rangle^2, \qquad \lambda > 0. \tag{30}$$

A nonzero vacuum requires the sign and magnitude of the quadratic term plus stabilization.

[NG] Pure positive Dirichlet energy does not select a nonzero amplitude.

12.1 Schur-complement dressing

For a block scalar quadratic form

$$\begin{pmatrix} A & B \\ B^\dagger & C \end{pmatrix}, \tag{31}$$

eliminating the hidden block gives

$$A_{\text{eff}} = A - BC^{-1}B^\dagger. \tag{32}$$

[C] Coupling to a hidden sector can lower the effective curvature of an already permitted scalar direction.

[NG] If the full quadratic form is positive, its Schur complement cannot become negative. Thus Feshbach reduction can transmit or reveal an instability; it cannot create spontaneous symmetry breaking from an everywhere-positive quadratic theory.

Historical specific scalar-mass and quartic claims are not retained.

13 Neutral fermions and neutrino mass structure

Write all neutral Weyl components in a common left-handed basis and consider

$$\mathcal{M}_\nu = \begin{pmatrix} M_L & M_D \\ M_D^T & M_R \end{pmatrix}, \qquad M_L^T = M_L, \quad M_R^T = M_R. \tag{33}$$

If M_R is invertible, the algebraic Schur complement is

$$M_{\text{Schur}} = M_L - M_D M_R^{-1} M_D^T. \tag{34}$$

In the usual heavy-gap type-I limit with $M_L = 0$,

$$M_{\text{light}} \approx -M_D M_R^{-1} M_D^T. \tag{35}$$

[D] The Schur algebra is exact.

[C] A seesaw interpretation is available if a physical heavy neutral sector and the necessary chiral/scalar intertwiners are independently supplied.

[NG] Current HSMT does not select among massless, Dirac, Majorana, pseudo-Dirac/mixed, or seesaw realizations.

[NG] A neighboring grade is not automatically a sterile or right-handed neutrino. Lepton-number status remains additional physical structure.

14 Relational gravity maturation

The gravity programme provides the clearest example of why a theorem-boundary analysis is necessary. The mature route is

$$\begin{aligned} \text{microscopic state} \longrightarrow \text{correlation geometry} \longrightarrow \text{collective geometry} \longrightarrow \text{response kernel}, \\ \longrightarrow \text{redundancy} \longrightarrow \text{Ward structure} \longrightarrow \text{tensor dynamics}. \end{aligned} \tag{36}$$

Several arrows can be realized in controlled benchmarks. The entire chain cannot yet be completed.

14.1 GC-0: emergent geometry is not gravity

A metric may determine distances, volumes, Laplacians, characteristic propagation, and curvature. It need not itself be dynamical.

[NG] emergent metric $\not\Rightarrow$ gravity. A genuine gravitational theory requires reciprocal matter backreaction and collective geometric dynamics.

14.2 GC-1: explicit dynamic geometry can still be elastic

A finite benchmark was constructed on

$$X_L = (\mathbb{Z}/L\mathbb{Z})^2 \tag{37}$$

with positive conductances $w_e = e^{q_e}$ and harmonic fast operator

$$A(q) = m_f^2 I + B^T W(q) B. \tag{38}$$

The fast ground-state covariance is

$$C(q) = \frac{1}{2} A(q)^{-1/2}, \tag{39}$$

so $A(q) = (2C(q))^{-2}$. This gives exact covariance-to-operator reconstruction in the declared benchmark.

A local co-metric may be reconstructed as

$$g_x^{ab} = \sum_r w_{xr} d_r^a d_r^b. \tag{40}$$

The geometry can therefore respond dynamically to sources, and a controlled Born–Oppenheimer slow regime can be constructed.

Nevertheless, variations of q change physical conductances and the microscopic spectrum. They are physical strains.

[NG] dynamic geometry $\not\Rightarrow$ gauge gravity.

14.3 GC-2: chart redundancy is real but insufficient

Introduce explicit reconstruction/chart data ζ,

$$g = R[C; \zeta]. \tag{41}$$

Pure chart variation may leave intrinsic physical data invariant while changing metric components by

$$\delta_\xi g = \mathcal{L}_\xi g. \tag{42}$$

The full $ME0$ package transforms consistently: metric components, measure density, scalar-field gradients, and energy-form representation all change while the intrinsic physical functional remains the same.

[D] Continuous vertical chart redundancy can therefore be constructed in the declared reconstruction model.

[NG] Any intrinsic theory can be parametrized this way. Hence general covariance does not imply gravitational gauge symmetry.

14.4 GC-3: microscopic relabelings can have a continuum limit

On the periodic support, define row-dependent permutations

$$P_L^f(i,j) = (i + s_j^{(L)}, j) \pmod{L}, \tag{43}$$

with $s_j^{(L)}/L \to a f(j/L)$. Then the coordinate limit is

$$F_f(x,y) = (x + a f(y), y), \tag{44}$$

whose Jacobian has determinant one. Thus microscopic permutations can approximate nontrivial local volume-preserving continuum reparametrizations. Analogous column shears generate a large divergence-free Lie algebra under brackets.

[C] A substantial SDiff-like continuum ancestry is therefore supported.

Theorem 14.1 (Measure preservation under permutation refinement)**.** *Let*

$$\mu_L = \frac{1}{N_L} \sum_{i=1}^{N_L} \delta_{x_i^{(L)}}$$

be uniform empirical measures converging weakly to μ. Let P_L be permutations of the support points and let the induced coordinate maps F_L converge uniformly to a continuous map F. Then

$$F_* \mu = \mu.$$

Proof. Every permutation satisfies $(F_L)_* \mu_L = \mu_L$. For every continuous test function φ,

$$\int \varphi \, d\mu_L = \int \varphi \circ F_L \, d\mu_L.$$

Taking the refinement limit and using weak convergence of μ_L together with uniform convergence $F_L \to F$ yields

$$\int \varphi \, d\mu = \int \varphi \circ F \, d\mu,$$

so $F_* \mu = \mu$. □

Corollary 14.2. *For a smooth limit with uniform coordinate volume measure, microscopic permutation ancestry is restricted to the measure-preserving continuum sector unless additional sampling/refinement structure enters.*

[NG] Theorem 14.1 supports an SDiff-like bound. It does not derive full Diff or gravity.

14.5 GC-4: intrinsic relational static action

The GC-1 benchmark originally included a declared slow stiffness $\frac{\tau}{2} q^T L_s q$, which carries a fixed local reference structure. A background-erasure benchmark removed it and retained the fast zero-point energy

$$E_0[q] = \frac{1}{2} \operatorname{Tr} \sqrt{A(q)}. \tag{45}$$

Under a microscopic relabeling, $A \to PAP^T$, and therefore E_0 is exactly invariant. This is a genuine model result.

Theorem 14.3 (Monotonicity of the inherited harmonic spectral energy)**.** *Let*

$$A(q) = m_f^2 I + \sum_e e^{q_e} b_e b_e^T$$

with $m_f > 0$, and define $E_0(q) = \frac{1}{2} \operatorname{Tr} \sqrt{A(q)}$. Then for every physical edge e,

$$\frac{\partial E_0}{\partial q_e} = \frac{1}{4} e^{q_e} b_e^T A(q)^{-1/2} b_e > 0.$$

Consequently E_0 has no finite unconstrained interior stationary point in the q-coordinates.

Proof. For positive A,

$$\delta \operatorname{Tr} \sqrt{A} = \frac{1}{2} \operatorname{Tr}(A^{-1/2} \delta A).$$

Since $\partial_{q_e} A = e^{q_e} b_e b_e^T$,

$$\partial_{q_e} E_0 = \frac{1}{4} e^{q_e} \operatorname{Tr}(A^{-1/2} b_e b_e^T) = \frac{1}{4} e^{q_e} b_e^T A^{-1/2} b_e.$$

Because $A^{-1/2}$ is positive definite and $b_e \neq 0$, the expression is strictly positive. □

[NG] The current inherited harmonic relational energy is background-free but vacuum-unstable unless new independently justified structure enters.

14.6 Geometry-locked measure

For a local reconstructed co-metric g^{ab}, define a candidate local volume weight

$$m_x(q) = [\det g_x^{ab}(q)]^{-1/2}, \tag{46}$$

and $M(q) = \operatorname{diag}(m_x(q))$. Then define

$$A_D(q) = m_f^2 I + M(q)^{-1/2} B^T W(q) B M(q)^{-1/2}. \tag{47}$$

This operator remains conjugation covariant under relabeling.

[C] A fixed independent physical volume form is therefore not mathematically necessary.

[NG] Under a common logarithmic conductance scaling the same uniform runaway persists. Geometry-locking the measure does not stabilize the overall geometry.

14.7 GC-5: dynamical determinant and induced inertia

The harmonic fast sector has Euclidean determinant

$$\Gamma_{\text{fast}}[q] = \frac{1}{2} \operatorname{Tr} \log \left(-\partial_\tau^2 + A(q) \right). \tag{48}$$

For static q, this reduces to the energy in (45).

The full harmonic determinant admits an explicit quadratic response kernel. In the real nondegenerate gapped branch its low-frequency expansion contains a positive-semidefinite induced inertia, no local Berry-curvature term, and is analytic below the first two-fast-quantum threshold. This demonstrates that static relational geometry and temporal collective response can arise from the same microscopic operator. Comparison of inequivalent lattice refinements with the same leading continuum principal symbol nevertheless shows that the bare response coefficients are regulator dependent. A physical renormalization or universality principle is therefore still required. The detailed kernel and regulator comparison are given in Appendices D and E.

A standard resolvent-decay mechanism provides context for the quasi-locality of gapped local operators [14].

14.8 Static relabeling is not spacetime gauge symmetry

For a time-independent permutation P,

$$-\partial_t^2 + A \longrightarrow P(-\partial_t^2 + A)P^T, \tag{49}$$

and the determinant is invariant. If $P = P(t)$, however,

$$\partial_t(P\phi) = P\dot{\phi} + \dot{P}\,\phi, \tag{50}$$

so temporal connection terms appear.

[NG] Time-independent relabeling symmetry is not yet a spacetime diffeomorphism gauge symmetry.

[OPEN] A continuum temporal gauge ancestry or independently derived temporal connection is missing.

Classic consistency arguments for massless spin-2 fields are relevant only after the corresponding gauge and source structure is present [15, 16]; HSMT has not yet reached that stage.

15 What the gravity programme has actually achieved

The negative closures should not obscure the positive results. The post-maturation gravity programme has established the following controlled facts.

1. **[D]** A finite covariance geometry can be explicitly reconstructed in a declared microscopic model.
2. **[C]** A dynamically controlled slow geometric manifold can exist.
3. **[D]** A geometric susceptibility can be calculated once the slow kernel is given.
4. **[D]**/**[C]** Discrete Lie-derivative kinematics can converge toward the continuum vector-field algebra.
5. **[D]**/**[C]** Locality-preserving microscopic permutations can possess continuous volume-preserving continuum ancestry.
6. **[D]** A fixed local reference stiffness is not mathematically required for a relational static geometric energy.
7. **[C]** The physical measure can be locked to the same dynamical geometry.
8. **[D]** The harmonic fast sector gives an explicit relational functional determinant in the declared model.
9. **[D]** Static stiffness and induced inertia can arise from the same microscopic operator in that benchmark.
10. **[D]** The gapped low-frequency response is analytic below the fast pair threshold.

These are nontrivial advances. They are not yet general relativity.

16 Gravity hard boundary

Current gravity boundary.

Background-free relational static geometry	Conditionally available.
Stable nontrivial relational vacuum	Not derived.
Regulator-universal continuum kernel	Not derived.
Spacetime Diff/Ward ancestry	Not derived.
Physical massless spin-2 sector	Not derived.

Consequently,

$$\text{GC-6 is not opened under the unchanged frozen core.} \tag{51}$$

A continuation of the gravity branch requires genuinely new microscopic physics.

17 Negative closures as scientific results

The maturation programme derives much of its value from failed selector searches that can now be closed rather than repeatedly revisited.

Table 3: Major selector failures and no-go results established by the post-maturation programme.

Closed statement	Mature conclusion
Complete exchangeability ⇒ physical locality	**[NG]** False without new locality data.
Grade ⇒ dimension	**[NG]** No universal map $N \mapsto d$.
Grade ⇒ time	**[NG]** Time requires independent $T0$.
Closed unitary dynamics ⇒ arrow	**[NG]** No generic monotone arrow.
Feshbach exactness ⇒ physical projector	**[NG]** Projection dynamics does not select P.
Exterior/Hodge Clifford algebra ⇒ physical fermions	**[NG]** Physical spinor/CAR sector requires additional structure.
Clifford commutant ⇒ Standard Model gauge group	**[NG]** The commutant gives a possibility space, not a selector.
K4/triality/cycle rank ⇒ three generations	**[NG]** No unique family-number selector.
Profile overlaps ⇒ flavor predictions	**[NG]** Free profiles are too expressive.
Grade/ρ/Gaussian width ⇒ Higgs field	**[NG]** No derived scalar order parameter.
Feshbach structure ⇒ neutrino branch	**[NG]** Dirac/Majorana/seesaw remain unselected.
Metric existence ⇒ gravity	**[NG]** Geometry can remain nondynamical.
Dynamic geometry ⇒ gravity	**[NG]** Elastic geometry is a counterexample.
Chart covariance ⇒ gravity	**[NG]** Parametrization is too weak.
Large permutation group ⇒ full Diff	**[NG]** Locality and measure restrictions matter.
Relabeling invariance ⇒ gravitational action	**[NG]** Many inequivalent invariant actions exist.

Closed statement	Mature conclusion
Background-free spectral energy ⇒ stable vacuum	[NG] Harmonic benchmark runs to zero coupling.
Principal-symbol matching ⇒ continuum universality	[NG] Bare response coefficients remain regulator dependent.
Static label symmetry ⇒ spacetime Ward identity	[NG] Time-dependent gauge ancestry is absent.

These are not ancillary disclaimers. They define the theory's present logical boundary.

18 Primitive and empirical inputs

The current architecture contains several data classes that are explicitly primitive or empirically supplied.

1. [P] $L0$: locality filtration.
2. [P] $ME0$: physical metric, measure, and energy normalization.
3. [P] $T0$: affine clock and temporal generator calibration.
4. [P] $O0/A0$: orientation and statistical-arrow data.
5. [P] $PB0$: operational identification of the physical sector.
6. [P]/[OPEN] Physical tangent/spinor module and fermionic quantization.
7. [P]/[OPEN] Physical internal gauge group, representations, connection, and couplings.
8. [P]/[OPEN] Family number.
9. [P]/[OPEN] Physical flavor-profile dynamics.
10. [P]/[OPEN] Scalar representation, potential, stabilization, and vacuum.
11. [P]/[OPEN] Neutral-fermion content and lepton-number structure.

Declaring these inputs explicitly is scientifically preferable to hiding them inside notation.

19 Genuine new-physics entry points

The current gravity branch has reached an input boundary. Future progress requires at least one genuinely new ingredient.

19.1 Intrinsic geometric stabilizer

[OPEN] A relational invariant could constrain the runaway scale and select a finite geometric vacuum. Examples worth testing only if independently motivated include a conserved relational volume, a topological invariant, a microscopic normalization law, or a gauge-generated constraint. No present theorem supplies one.

19.2 Interacting fast sector

The harmonic fast sector gives a monotone zero-point energy. **[OPEN]** Interactions or many-body effects could generate competing intrinsic terms. Such terms would be legitimate only if derived from microscopic dynamics, not selected because their continuum form resembles general relativity.

19.3 Topology and refinement dynamics

[OPEN] Dynamical topology or refinement could both stabilize geometry and reduce regulator dependence.

19.4 Measure/refinement co-dynamics

[OPEN] A law in which sampling density, physical measure, and reconstruction co-evolve with geometry may enlarge the current SDiff-like ancestry.

19.5 Temporal gauge ancestry

[OPEN] A microscopic temporal connection or constraint may be required for a spacetime-dependent redundancy.

19.6 Universality / renormalization

[OPEN] A physical continuum requires an explanation of why distinct microscopic regulators flow to the same normalized infrared response. Possibilities include an RG fixed point, a universality class, or a spectral renormalization theorem. None is presently frozen.

20 Historical claims and mature status

Table 4: Historical HSMT targets and their mature epistemic status.

Historical target	Mature status	Reason
Physical $3+1$ spacetime	**[NG]** Not derived	Grade and dimension remain distinct; no physical dimensional selector is frozen.
Specific physical spectral dimension	**[NG]** Not derived	Different support/ME0 realizations produce different sectoral exponents.
Standard Model gauge group	**[NG]** Not derived	Clifford commutant supplies possibilities but no unique physical selection.
Three generations	**[NG]** Not derived	Family multiplicity has no internal selector.
Observed mass ratios	**[NG]** Not derived	Flavor profiles and kernels remain underdetermined.

Historical target	Mature status	Reason
CKM/PMNS matrices	[NG] Not derived	Sector mixing requires missing physical flavor dynamics.
Higgs representation	[NG] Not derived	Scalar kinematics does not select internal representation.
Higgs vacuum / mass / quartic	[DEMOTED]	No derived physical scalar potential or vacuum.
Dirac versus Majorana neutrinos	[NG] Not derived	Lepton-number and heavy-neutral content remain unresolved.
Seesaw scale	[NG] Not derived	Physical heavy neutral sector not selected.
Einstein equations	[NG] Not derived	No stable regulator-universal Ward-null tensor continuum.
Newton constant	[NG] Not derived	No normalized physical spin-2 kinetic/source sector exists.
Small cosmological constant	[DEMOTED]	Raw spectral vacuum energy is refinement extensive; leakage suppression is not a derivation.
Black-hole solutions	[DEMOTED]	Nonlinear gravitational field equations have not been derived.
Gravitational-wave dispersion	[DEMOTED]	No physical tensor kernel has been isolated.
Dark matter / dark energy from shell leakage	[DEMOTED] / hypothesis	No baseline physical gravitational response has been derived.
Historical 1/12 chain	[C] Conditional only	Its microscopic selector remains absent.

21 The current maturation graph

Graded arena

↓

Associative operator laboratory

↓

Projection / Feshbach dynamics

↓

$L0 + ME0 + T0$

↓

Conditional local spacetime structure

↓

Clifford/fermionic capacity; internal-symmetry possibility space; family/flavor structures; scalar/neutral-mass mechanisms; relational gravity route

↓

Explicit new-input boundary

This endpoint is itself a major maturation result.

22 Discussion

The strongest change produced by the post-maturation programme is not the addition of another speculative physical sector. It is the systematic separation of five logically different statements:

1. a mathematical structure exists;
2. the structure can host a physical mechanism;
3. the mechanism follows conditionally from additional assumptions;
4. the theory selects the required physical input;
5. Nature realizes that selected branch.

Earlier unification programmes often move too quickly from item 1 to item 5. The present HSMT theorem boundary deliberately does not. The resulting framework is less phenomenologically complete than the earliest HSMT proposals, but its logical dependencies and failure conditions are substantially more explicit.

For example, the gravity programme no longer relies on the statement that curvature exists and therefore gravity follows. It demands

$$\text{geometry} + \text{dynamics} + \text{redundancy} + \text{backreaction} + \text{continuum universality}. \tag{52}$$

Likewise, the existence of a Clifford module no longer licenses the claim that physical fermions or the Standard Model follow. The existence of a family multiplicity space no longer licenses three generations. A Gaussian overlap no longer counts as a flavor prediction unless profile dynamics is derived.

These restrictions are productive. They identify the exact points at which future microscopic physics must enter.

23 Conclusion

HSMT has matured from a broad unification proposal into a controlled hierarchical operator framework with explicit locality, geometric, matter-sector, flavor, scalar, neutrino, and gravity closure analyses.

Several exact and conditional structures survive: the graded operator architecture; exact projection and Feshbach reduction; relational support/Hodge constructions; a dimension-neutral locality and metric-measure-energy framework; Clifford and commutant algebraic capacity; family and flavor carrier structures; scalar and neutral-mass conditional mechanisms; locality-preserving continuum relabeling constructions; an intrinsic relabeling-invariant spectral geometric energy; and a gapped microscopic determinant with a controlled analytic low-frequency kernel and positive induced collective inertia.

The same maturation programme establishes equally important nonselection results. Current HSMT does not uniquely derive the observed spacetime dimension, the physical grade, the observed projector, the Standard Model gauge structure, three fermion generations, observed flavor matrices, the physical Higgs sector, the neutrino mass branch, a stable regulator-universal gravitational continuum, or Einstein gravity.

The missing selectors are no longer hidden inside broad terminology. They are localized as explicit new-physics interfaces.

Current maturity statement.
HSMT presently provides a graded operator architecture, controlled projection dynamics, explicit locality and metric-measure-energy layers, several conditional matter-sector mechanisms, and a relational route toward dynamical geometry. Its maturation programme has also established multiple nonselection and no-go results. Current results do not uniquely derive the observed spacetime dimension, Standard Model content, symmetry-breaking vacuum, neutrino mass branch, or Einstein gravity. These missing selectors are now localized as explicit new-physics interfaces rather than hidden assumptions.

The appropriate next scientific move is therefore not another selector search inside the unchanged frozen core. Future progress requires either consolidation and publication of the present theorem boundary or introduction of one independently motivated microscopic ingredient followed by the same adversarial closure discipline.

The central methodological conclusion is

$$\begin{gathered}\text{derived structure} \neq \text{physical selection},\\ \text{mathematical capacity} \neq \text{physical realization},\\ \text{conditional mechanism} \neq \text{prediction}.\end{gathered} \tag{53}$$

That distinction is the current scientific spine of matured HSMT.

A Claim-status summary

Table 5: Consolidated claim-status summary.

Topic	Status	Mature statement
Graded arena	**[D]**	Integer hierarchy architecture retained.
Absolute grade origin	**[D]**/**[NG]**	Additive origin is conventional; no absolute selector.
Physical dimension	**[OPEN]**/**[NG]**	No value presently derived.
Locality	**[P]**	$L0$ required; full exchangeability does not select it.
Metric-measure-energy	**[P]**	$ME0$ required for cardinal physical geometry.
Feshbach reduction	**[D]**	Exact once projector supplied.
Physical projector	**[P]**/**[OPEN]**	Not selected by Feshbach or isolation.
Time	**[P]**	$T0$ supplied independently of grade.
Arrow	**[P]**/**[NG]**	Not generated generically by reversible unitary dynamics.
Clifford seed	**[D]**/**[C]**	Available given physical metric/cochain structure.
Physical fermions	**[OPEN]**	Require additional module/global/CAR structure.

Topic	Status	Mature statement
Internal symmetry space	[D]	Clifford commutant identifies possibilities.
Physical gauge group	[OPEN]/[NG]	Not selected.
Family multiplicity	[D]	Family-space carrier identified.
Three generations	[NG]	No selector.
Flavor overlap	[C]	Mechanism exists but free-profile universality blocks prediction.
Scalar kinematics	[D]	0-forms/0-cochains provide scalar carrier.
Physical Higgs sector	[OPEN]/[NG]	Potential and vacuum not derived.
Neutral mass Schur structure	[D]	Algebraically exact.
Physical neutrino branch	[OPEN]/[NG]	Not selected.
Relational static gravity action	[D]/[C]	Inherited harmonic spectral term exists in the declared model.
Stable relational vacuum	[NG]	Not obtained in current harmonic branch.
Induced inertia	[D]/[C]	Derived fast-sector contribution in the declared model.
Regulator-universal gravity kernel	[OPEN]/[NG]	Bare coefficients not universal.
Spacetime Ward ancestry	[OPEN]/[NG]	Static relabeling does not provide it.
Massless graviton	[NG]	Not derived.
Einstein equations	[NG]	Not derived.

B Frozen historical firewalls

The following historical routes remain outside the current derived result set unless genuinely new microscopic information enters:

- a unique σ_0 selector from the frozen core;
- revival of the conditional 1/12 spectral chain as a derived prediction;
- WKB reparameterization used as a microscopic selector;
- $x = 16$ or $\omega a \rho^2 = 16$ used as a fundamental selector without an independent derivation;
- three-family selection from multiplicity/cycle/triality numerology;
- post-hoc identification of a grade with the observed physical dimension;
- post-hoc identification of hierarchy leakage with dark matter, dark energy, or the cosmological constant.

C New-input boundary for gravity

To reopen the gravity branch beyond GC-5, at least one genuinely new microscopic ingredient is required.

Required conjunction for a reopened physical gravity branch.
A successful extension would need all of the following:

1. a stable nontrivial relational vacuum;
2. a regulator-controlled continuum response;
3. microscopically ancestral spacetime redundancy;
4. physical Ward/null structure;
5. healthy tensor dynamics.

No present frozen HSMT structure supplies that conjunction.

D Harmonic determinant and low-frequency response

The gapped harmonic fast action is

$$S_{\text{fast}} = \frac{1}{2}\int d\tau \left[\dot{\phi}^T\dot{\phi} + \phi^T A(q)\phi\right]. \tag{54}$$

Integrating out ϕ yields (48). Let

$$A\,|n\rangle = \omega_n^2\,|n\rangle\,, \tag{55}$$

and define

$$A_A = \partial_A A, \qquad A_{AB} = \partial_A\partial_B A. \tag{56}$$

The Euclidean quadratic response is

$$K_{AB}(i\Omega) = \frac{1}{4}\sum_n \frac{(A_{AB})_{nn}}{\omega_n} \tag{57}$$

$$- \sum_{n,m} \frac{(A_A)_{nm}(A_B)_{mn}(\omega_n+\omega_m)}{4\omega_n\omega_m\left[(\omega_n+\omega_m)^2+\Omega^2\right]}. \tag{58}$$

At small Ω,

$$K_{AB}(i\Omega) = K_{AB}^{(0)} + \Omega^2 M_{AB}^{(\text{ind})} + O(\Omega^4), \tag{59}$$

where

$$M_{AB}^{(\text{ind})} = \sum_{n,m} \frac{(A_A)_{nm}(A_B)_{mn}}{4\omega_n\omega_m(\omega_n+\omega_m)^3}. \tag{60}$$

For $A = B$, $M_{AA}^{(\text{ind})} \geq 0$. In the real nondegenerate branch the ground-state Gaussian can be chosen locally real, so the local Berry curvature vanishes. Pair thresholds begin at

$$\omega_n + \omega_m \geq 2m_f. \tag{61}$$

Thus the low-frequency derivative expansion is analytic while the fast gap remains uniformly open.

This is a derived response in the declared harmonic benchmark. It is not a physical graviton propagator.

E Regulator comparison and continuum-universality test

Consider two inequivalent local refinement complexes with the same leading small-momentum principal symbol.

Regulator S:

$$\lambda_S(k) = 4\sin^2\frac{k_x}{2} + 4\sin^2\frac{k_y}{2} = k_x^2 + k_y^2 + O(k^4). \tag{62}$$

Regulator D:

$$\lambda_D(k) = 2\sin^2\frac{k_x}{2} + 2\sin^2\frac{k_y}{2} + \sin^2\frac{k_x + k_y}{2} + \sin^2\frac{k_x - k_y}{2} = k_x^2 + k_y^2 + O(k^4). \tag{63}$$

The benchmark large-refinement coefficients in declared microscopic units are

Quantity	S	D
E_0/N	1.4026289011	1.3181647553
$N^{-1}dE_0/dq$	0.3384784240	0.2766986333
$K(0)/N$	0.2494922748	0.2145384391
c in $K(p) = K(0) + cp^2$	0.0129817170	0.0119990829
$M_{\mathrm{ind}}(0)/N$	0.00251465860	0.00209070789

These numbers are benchmark quantities, not physical predictions. Their role is to falsify the proposition that matching the leading continuum principal symbol is sufficient to produce universal bare response coefficients. A state-independent vacuum subtraction cannot remove differences in the q-dependent force, Hessian, gradient coefficient, or induced inertia. A renormalization/universality prescription remains missing.

F Relationship to earlier HSMT papers

The present paper is intended to be read as a post-maturation companion to the earlier HSMT papers.

The September 2026 reference framework remains the conceptual public spine. It records the graded arena, associative laboratory, projection architecture, and downstream extension programme. The subsequent Paper 2 develops inverse-spectral reconstruction and identifiability/no-go boundaries for the reduced hierarchy. The present paper does not modify their historical content. It places both inside a stricter epistemic classification and records the closures obtained afterward.

A proposed derived-dimension Paper 3 remains premature because no physical dimension has crossed the theorem boundary.

References

[1] V. M. Garrett, *Hierarchical Shell-Manifold Theory: A Matured Foundational and Reference-Model Framework*, HSMT manuscript, September 2026.

[2] V. M. Garrett, *Hierarchical Shell Manifold Theory: Operator Reconstruction, Spectral Identifiability, and No-Go Boundaries*, HSMT manuscript, 2026.

[3] T. Kato, *Perturbation Theory for Linear Operators*, 2nd ed., Springer-Verlag, Berlin, 1976; reprinted in Classics in Mathematics, Springer, 1995.

[4] H. Feshbach, "Unified theory of nuclear reactions," *Annals of Physics* **5**, 357–390 (1958), doi:10.1016/0003-4916(58)90007-1.

[5] H. Feshbach, "A unified theory of nuclear reactions. II," *Annals of Physics* **19**, 287–313 (1962), doi:10.1016/0003-4916(62)90221-X.

[6] S. Nakajima, "On Quantum Theory of Transport Phenomena: Steady Diffusion," *Progress of Theoretical Physics* **20**, 948–959 (1958), doi:10.1143/PTP.20.948.

[7] R. Zwanzig, "Ensemble Method in the Theory of Irreversibility," *Journal of Chemical Physics* **33**, 1338–1341 (1960), doi:10.1063/1.1731409.

[8] H.-P. Breuer and F. Petruccione, *The Theory of Open Quantum Systems*, Oxford University Press, Oxford, 2007.

[9] A. Connes, *Noncommutative Geometry*, Academic Press, San Diego, 1994.

[10] M. Fukushima, Y. Oshima, and M. Takeda, *Dirichlet Forms and Symmetric Markov Processes*, De Gruyter Studies in Mathematics, Vol. 19, Walter de Gruyter, Berlin, 1994.

[11] F. R. K. Chung, *Spectral Graph Theory*, CBMS Regional Conference Series in Mathematics, Vol. 92, American Mathematical Society, Providence, 1997.

[12] H. B. Lawson, Jr. and M.-L. Michelsohn, *Spin Geometry*, Princeton Mathematical Series, Vol. 38, Princeton University Press, Princeton, 1989.

[13] G. Calcagni, "Multifractional theories: an unconventional review," *Journal of High Energy Physics* **2017**, 138 (2017), doi:10.1007/JHEP03(2017)138.

[14] J. M. Combes and L. Thomas, "Asymptotic behaviour of eigenfunctions for multiparticle Schrödinger operators," *Communications in Mathematical Physics* **34**, 251–270 (1973), doi:10.1007/BF01646473.

[15] S. Weinberg, "Photons and Gravitons in S-Matrix Theory: Derivation of Charge Conservation and Equality of Gravitational and Inertial Mass," *Physical Review* **135**, B1049–B1056 (1964), doi:10.1103/PhysRev.135.B1049.

[16] S. Deser, "Self-interaction and gauge invariance," *General Relativity and Gravitation* **1**, 9–18 (1970).